Testing a spectral epidemic threshold for SIS on networks under heavy-tailed recovery

Jan Ruijgrok
Open University of the Netherlands

April 2026

Abstract

Many infectious diseases do not confer lasting immunity, so infections may die out quickly or persist at low prevalence. Network SIS models often summarise "control" using compact rules that link spreading strength, mean infectious period, and connectivity—notably via the leading eigenvalue of the contact structure. Real recovery times can be irregular and right-skewed (*heavy-tailed* in the sense of substantial mass in long infectious periods) even when their average matches a simple reference case; our research asks whether such spectral mean-field thresholds still track outcomes in an explicit agent-based model. We implemented a group-recovery (grp) SIS formulation on Erdős–Rényi contact networks (N=2000, mean degree six), exporting edges per graph draw to obtain the spectral reference numerically. Every run starts from I_0=5 infected agents (ρ_0=0.25%). On three independent draws (`expt-seed` 10001–10003) we ran replicated sweeps of infection strength under exponential, Tang power law (λ_{PL}=4.24), and Tang lognormal ($\sigma_{\ln}$=1) recovery laws with the same mean infectious duration $\mathbb{E}[W]$=5 ticks; experiment `08` adds per-tick output comparing microscopic prevalence to the homogeneous mean-field curve in the same implementation when extinction does not end runs early. Infection strengths at which half of stochastic replicates still carry the pathogen lie systematically above the mean-field prediction; differences between heavy-tailed and Markovian regimes in where that threshold falls are modest compared to this gap, whereas extinction timing depends more clearly on the recovery law. Overall, the spectral line is a useful conservative guide rather than a sharp critical value here, and heavy tails mainly reshape variability and transients more than they uniformly lower the survival threshold, partly qualifying the hypothesis that long tails alone strongly ease persistence at an unchanged average recovery duration.

Keywords: network epidemiology; SIS model; heavy-tailed recovery; agent-based simulation; spectral threshold; Erdős–Rényi random graph.

Notation and symbols

Table 1: main mathematical symbols.
Table 2: selected NetLogo/BehaviorSpace output names.

Table 1: Mathematical notation (discrete-time SIS on an unweighted simple graph unless noted).

Symbol	**Meaning**
N	Number of nodes (agents).
A, A_{ij}	Adjacency matrix; A_{ij}=1 if nodes i, j are neighbours (unweighted graph).
$\lambda_{\max}$	Spectral radius (largest eigenvalue of A) for the exported contact network.
W	Random infectious period (recovery time) at the agent level; $\mathbb{E}[W]$ is its mean, fixed across recovery regimes in our comparisons.
β	Per-tick Bernoulli probability of transmission along an edge from an infected agent (NetLogo *infection-prob*); one model tick is one discrete time step.
τ	Effective transmission intensity; we use $\tau = \beta\,\mathbb{E}[W]$ (mean per-tick success probability times mean infectious duration in ticks). This matches the scale-free grouping used with continuous-time mean-field thresholds when $\mathbb{E}[W]$ is expressed in the same time units as β (Section 2).
τ_c	Critical value of τ in spectral/mean-field theory; often $\tau_c \approx c/\lambda_{\max}$.
c	Dimensionless prefactor in threshold scaling (mean-field benchmark c=1).
β_{pred}	Spectral prediction $\beta_{\text{pred}} = 1/(\lambda_{\max}\mathbb{E}[W])$ at c=1.
τ_{pred}	$\tau_{\text{pred}} = \beta_{\text{pred}}\mathbb{E}[W] = 1/\lambda_{\max}$.
$\hat{\beta}_{\text{surv}\,50}$	Smallest β on the sweep grid for which at least 50% of replicates are not extinct by the time limit ("survival" threshold).
$\hat{\tau}_{50}$	$\hat{\beta}_{\text{surv}\,50}\,\mathbb{E}[W]$.
ρ	Fraction infected (prevalence among S+I); microscopic vs. mean-field curves in experiment `08`.
S, I	Susceptible and infected compartments (SIS).
I_0	Number of infected agents at t=0 (`initial-infected` in NetLogo); here I_0=5 unless noted.
ρ_0	Initial prevalence I_0/N (fraction of agents infected at t=0).

Table 2: NetLogo/BehaviorSpace metrics referenced in this report (names follow CSV export conventions).

Name	**Meaning**
`bs-out-extinct`	Run ended in absorption with no infected agents (1) vs. infection still present at time limit (0).
`bs-out-final-tick`	Tick at run termination.
`bs-out-late-mean-prevalence`	Mean prevalence over a late time window (used in auxiliary persistence checks).
`bs-out-prev-grp` / `bs-out-prev-mf`	Microscopic grp-SIS prevalence vs. homogeneous mean-field SIS prevalence (experiment `08`).

1 Introduction

Many real-world infections do not give permanent immunity. People can be infected, recover, and then become infected again, sometimes multiple times in their life. Examples include common colds, certain sexually transmitted infections, and hospital-acquired infections. A central practical question is: *when does an introduced infection die out quickly, and when can it keep circulating in a population for a long time?* This boundary between die-out and long-term persistence is often called the epidemic (or endemic) threshold.

In a basic SIS (susceptible–infected–susceptible) model on a network, each individual is either healthy and susceptible or currently infected, and recovery is typically assumed to follow a simple, memoryless process. A standard summary of how easily infection can take hold is an *effective transmission* intensity τ that combines per-contact infection strength with the mean infectious period. Spectral arguments then predict an epidemic (endemic) threshold of the form

$$\tau_c \approx \frac{c}{\lambda_{\max}},$$

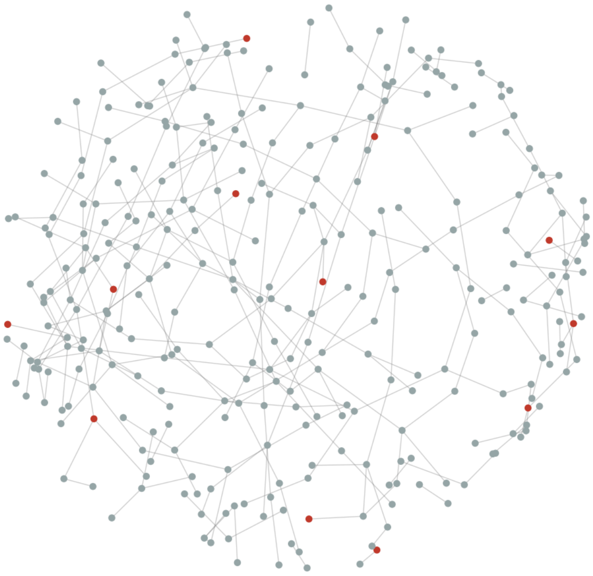

Figure 1: Induced subgraph on a random subset of nodes from the exported ER edge list (seed 10001), spring layout for visualisation only; full simulations use all 2000 nodes. Highlighted nodes illustrate a possible infected set.

where $\lambda_{\max}$ is the largest eigenvalue of the adjacency matrix and c is a constant. In a homogeneous mean-field limit one often takes $c = 1$; on a finite network, in discrete time, and with correlations beyond mean field, the effective c need not be exactly one, but the scaling with $1/\lambda_{\max}$ remains the usual reference. Equivalently, at fixed mean recovery time $\mathbb{E}[W]$, the implied critical *infection probability* is $\beta_c \approx \tau_c/\mathbb{E}[W] \approx c/(\lambda_{\max}\mathbb{E}[W])$. This spectral picture has been widely used and analysed in network epidemiology [3, 6, 4].

However, real recovery times are often far from this idealised picture. Tang et al. introduce the grp-SIS framework as a generalisation of the basic SIS model, allowing more flexible recovery-time patterns for infected individuals, including heavy-tailed distributions, while keeping the same basic SIS structure of possible reinfection [5]. Together with other work on SIS models with general or heavy-tailed recovery [2], this confirms that the *shape* of the recovery-time distribution can strongly influence dynamics, even when the *mean* recovery time is kept fixed.

What is much less clear from existing theory is whether the simple scaling $\tau_c \approx c/\lambda_{\max}$ (and the mean-field benchmark $c = 1$) still describes the threshold well once we move away from exponential recovery towards more realistic, heavy-tailed recovery times. This question matters for practice because such rules of thumb are sometimes used to judge whether an infection is "under control" on a given contact structure. Agent-based modelling (ABM) offers a natural way to explore this gap: by simulating many individuals on an explicit contact network, with realistic recovery-time distributions, we can directly observe whether infections die out or persist under different parameter settings, without relying only on mean-field approximations (i.e., replacing detailed neighbour-to-neighbour correlations with average infection levels in the population).

In this project we therefore use a NetLogo agent-based SIS model on a simple random network (Erdős–Rényi [1] with about 2000 nodes and average degree around 6) as a numerical "laboratory" (see Figure 1 for an example network).

Within this setting we compare exponential (Markovian) recovery to two Tang et al. laws with higher variance at the same mean: shifted Pareto / power law with shape parameter

λ_{PL}=4.24 (NetLogo `power-law-lambda`) and lognormal with $\sigma_{\ln}$=1 (`lognormal-sigma`), see Figure 2, while keeping $\mathbb{E}[W]$=5 ticks. This allows us to test how robust the spectral scaling $\tau_c \approx c/\lambda_{\max}$ remains in an explicit agent-based model and to see whether these alternatives mainly shift the epidemic threshold or primarily affect extinction times and variability.

2 Methods

Problem and research questions

Our hypothesis is that recovery laws with long right tails (Tang power-law and lognormal at fixed $\mathbb{E}[W]$ and fixed shape parameters as above) make it easier for the infection to persist—operationally, to meet the finite-horizon survival criterion in Section 2—than exponential recovery at the same mean infectious period. A related question is whether they also pull the empirical threshold toward the mean-field spectral line $\tau_c \approx 1/\lambda_{\max}$ (c=1), i.e. whether $\hat{\tau}_{50}/\tau_{\mathrm{pred}}$ moves closer to one than under exponential recovery; Section 4 revisits both parts with the estimates in Section 3.

We address the following research questions to test this hypothesis in the agent-based model on Erdős–Rényi networks with fixed parameters (about 2000 nodes and average degree around 6):

- **RQ1**. To what extent does the empirically observed threshold—in *infection probability* or in τ (*infection probability*$\times$*mean recovery time*)—agree with the spectral reference $\tau_c \approx c/\lambda_{\max}$ under the mean-field benchmark $c = 1$, for an exponential (Markovian) recovery distribution?
- **RQ2**. How does the empirical threshold shift when the recovery distribution is replaced by Tang et al.'s power law or lognormal recovery, while keeping $\mathbb{E}[W]$ fixed?
- **RQ3**. On this Erdős–Rényi topology (fixed *number of nodes* and *average degree*), does recovery under the Tang power-law or lognormal laws lead to a systematically different empirical threshold (in terms of infection probability) compared to exponential recovery, or are differences mainly visible in extinction-time distributions and prevalence variability?

Operationalisation (summary). Each research question is tied to simulation outputs in `output/raw/` and aggregated CSVs: **RQ1** compares $\hat{\beta}_{\mathrm{surv}\,50}$ and $\hat{\tau}_{50}$ to β_{pred} and τ_{pred} under exponential recovery (same I_0, graph, and horizon as other laws). **RQ2** repeats the comparison for power-law and lognormal regimes at identical $\mathbb{E}[W]$, $\lambda_{\max}$, and sweep design. **RQ3** contrasts not only $\hat{\beta}_{\mathrm{surv}\,50}$ but also `bs-out-final-tick` and prevalence series (experiments `05`–`08`) to separate threshold location from transient and variability effects.

Research setup

- **Simulated world**. Individuals are nodes on an Erdős–Rényi graph [1] generated in NetLogo with about *number of nodes* = 2000 and *average degree* = 6. Experiments `05`–`07` enumerate three independent draws via `expt-seed` $\in \{10001, 10002, 10003\}$ (with matching `output/edges/edges_ER_<seed>.csv` from experiment `01`); for each draw, exponential, power-law, and lognormal recovery use the *same* contacts. BehaviorSpace `repetitions` supply process noise conditional on each graph.
- **Initial condition**. At t=0, exactly I_0=5 agents are infected (`initial-infected` = 5), so $\rho_0 = I_0/N = 0.25\%$. This seeds a small local outbreak rather than a high-prevalence state, so the estimated survival thresholds are not dominated by a deliberately large initial reservoir of infection. Finite-N SIS dynamics can still depend on I_0 (e.g. early extinction risk); we hold I_0 fixed across recovery laws and do not perform an I_0 sensitivity sweep in this report.

- **Infection process**. A standard agent-based SIS process runs on each such network: infection can spread along links; after recovery an individual can be infected again. The NetLogo model sets *infection probability* (strength of transmission per time step along a link).
- **What we vary**. (i) The infection strength *infection probability*, swept over many values. (ii) The recovery time pattern: first a simple "memoryless" exponential pattern, then two heavy-tailed options implemented as in Tang et al. (power law and lognormal).
- **What we keep fixed**. The graph, the generative specification (ER, *number of nodes*, *average degree*), and the *mean recovery time*, so comparisons between recovery regimes are fair at the same $\mathbb{E}[W]$.
- **Reference prediction from the network**. The contact list is exported to a CSV file. In Python (3.x; numerical work uses the project scripts under `scripts/`), we compute the largest eigenvalue $\lambda_{\max}$ of the adjacency matrix. The mean-field spectral reference is $\tau_c \approx 1/\lambda_{\max}$ (i.e. $c = 1$ in $\tau_c \approx c/\lambda_{\max}$). At fixed mean recovery time $\mathbb{E}[W]$, the corresponding critical *infection probability* is $\beta_{\text{pred}} = 1/(\lambda_{\max}\, \mathbb{E}[W])$, which marks the spectral prediction on the β axis.

Measurement, empirical threshold rule, and estimation protocol

- **Run outcome**. Each BehaviorSpace run ends when an infection-free absorbing state is reached or when simulated time reaches a configured cap (`max-ticks`, here 10,000 for experiments `05`–`07`). We record whether the run is classified extinct (`bs-out-extinct` = 1), the final tick, and late-window mean prevalence (`bs-out-late-mean-prevalence`) for auxiliary persistence checks. Simulations use NetLogo 7.0.3 (BehaviorSpace table exports in `output/raw/`).
- **Persistence vs. extinction at the horizon**. A replicate counts as *surviving* (still persistent at the horizon) if `bs-out-extinct` = 0, i.e. at least one infected agent remains when the run stops at `max-ticks` or absorption; otherwise it is *extinct*. This replaces informal wording such as "typically stays present": persistence is **binary** per replicate at the configured tick cap (`max-ticks` = 10,000 for experiments `05`–`07`), **not** a quasi-stationary prevalence level. The 50% survival threshold $\hat{\beta}_{\text{surv}\,50}$ is the smallest swept β at which at least half of the 24 replicates survive under this rule.
- **Design size and sweep schedule**. For each recovery regime, experiments `05`–`07` use `repetitions` = 24 per (β, regime, `expt-seed`) and a uniform stepped sweep of `infection-prob` from 0.018 to 0.048 in steps of 0.002 (16 values). With three independent ER draws per law this is $24 \times 16 \times 3 = 1{,}152$ stochastic runs per recovery law unless runs fail validation. The same schedule is applied in all three recovery laws so that differences are not confounded by replication count or seed list.
- **Grid-based 50% survival threshold $\hat{\beta}_{\text{surv}\,50}$**. For each β on the grid, the *survival rate* is the fraction of replicates with `bs-out-extinct` < 0.5. We define $\hat{\beta}_{\text{surv}\,50}$ as the *smallest* β on the grid at which this fraction is at least 50%, implemented in `scripts/threshold_estimators.py`. If the curve never reaches 50% below the largest β, the estimator equals the largest grid point (right-censoring). The sweep uses a *uniform* step ($\Delta\beta = 0.002$); a coarse-to-fine refinement around an estimated crossing (or a logistic dose–response fit to survival vs. β) would reduce discretisation error when the crossing lies between grid points or at the boundary. Stratified bootstrap confidence intervals (n=400 resamples over the β grid, stratifying by β) use the same rule.
- **Uncertainty reporting**. Bootstrap intervals reflect stochastic replicate noise conditional on a fixed `expt-seed`. Cross-seed variation in $\hat{\beta}_{\text{surv}\,50}$ (and in extinction-time curves) is

summarised separately by comparing the three ER instances; `output/threshold_ratio_by_run.csv` lists one row per (recovery regime, seed).

- **Effective transmission**. We report $\hat{\tau}_{50} = \hat{\beta}_{\mathrm{surv}\,50}\,\mathbb{E}[W]$ and compare $\hat{\tau}_{50}/\tau_{\mathrm{pred}}$ to the mean-field benchmark c=1.
- **Link to RQ1–RQ3**. **RQ1**: agreement between $\hat{\beta}_{\mathrm{surv}\,50}$ and β_{pred} under exponential recovery. **RQ2**: shifts under power-law and lognormal recovery at fixed $\mathbb{E}[W]$. **RQ3**: extinction-time and variability differences even when $\hat{\beta}_{\mathrm{surv}\,50}$ is similar.

Discrete-time infection probability and continuous-time τ. In the NetLogo implementation, one *tick* is one discrete time step; *infection-prob* is the Bernoulli probability that an infected agent transmits to a given susceptible neighbour along an existing edge during that tick (see `netlogo/virus_simulation.nlogox`). Continuous-time network SIS theory often expresses thresholds through pairwise transmission and recovery *rates* [6, 4]. With time step $\Delta t = 1$ in tick units, a common small-probability link is $\beta \approx 1 - e^{-\nu\Delta t}$ for an underlying per-edge rate ν; for the β values here, β is therefore close to such a ν in numerical magnitude, but the simulation remains genuinely discrete. Holding the mean infectious duration $\mathbb{E}[W]$ in ticks fixed across recovery laws, we report $\tau = \beta\,\mathbb{E}[W]$ as the discrete analogue of "per-contact success per tick $\times$ mean infectious ticks" used alongside the spectral reference $\tau_{\mathrm{pred}} = 1/\lambda_{\max}$. Comparisons to $\tau_c \approx 1/\lambda_{\max}$ should be read as *benchmarking* a continuous-time mean-field scale against a discrete microscopic model, not as claiming equality with a specific continuous-time critical point.

grp-SIS recovery times in the agent-based model

Each infected agent draws an integer recovery deadline $W \geq 1$ (ticks) from `draw-recovery-time` at infection; the agent recovers when the infection clock reaches that deadline (`recovery-step` in `netlogo/virus_simulation.nlogox`). Exponential draws use a geometric waiting time with mean `recovery-mean` = 5. Lognormal (Tang) draws use `lognormal-sigma` = 1 with scale chosen so the theoretical mean matches `recovery-mean`. Power law (Tang) draws use shifted Pareto shape `power-law-lambda` = 4.24 with scale set for the same mean via the Tang parameterisation. All draws are rounded to integer ticks and truncated below at one tick, so $\mathbb{E}[W]$ in the simulation matches the intended mean up to discretisation; the two Tang laws have strictly larger variance than the exponential reference at this common mean (Figure 2).

Procedure (per new infection). `draw-recovery-time` in NetLogo: read regime from the chooser; if exponential, sample a geometric duration with the configured mean; if lognormal (Tang), sample from the lognormal with fixed σ and scale set so the theoretical mean matches `recovery-mean`; if power law (Tang), sample from the shifted Pareto with fixed shape and scale set for the same mean; round to integer ticks, replace any value below 1 by 1; store the draw as the agent's recovery countdown and decrement it once per tick until recovery. The BehaviorSpace sweep holds `recovery-mean` and regime-specific shape parameters fixed while stepping `infection-prob`; each (β, regime) cell uses `repetitions` = 24 independent stochastic replicates (estimation protocol above).

Software and reproducibility

NetLogo 7 BehaviorSpace experiments are defined in `netlogo/behaviorspace_experiments.xml` (mirrored in the model file). From the project root:

- `python scripts/run_pipeline.py --all` runs the ER-focused set: `01` (export `output/edges/edges_ER_<seed>.csv` only), threshold experiments `05–07`, and `08-dynamics-trajectory-ER`

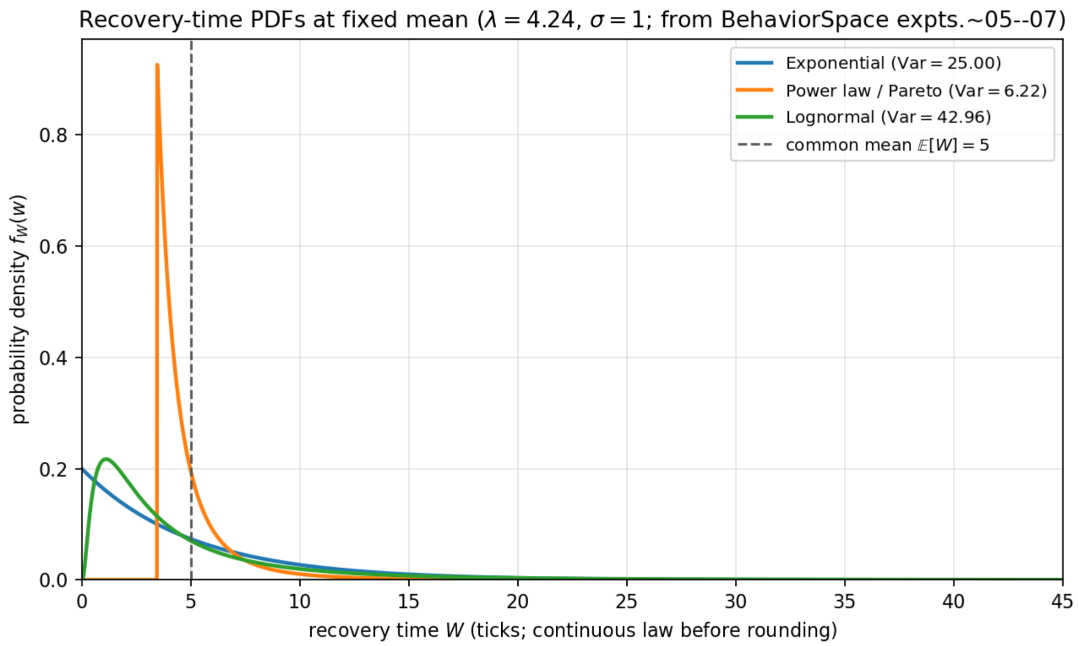

Figure 2: Probability density functions of the *continuous* recovery laws corresponding to the three regimes in BehaviorSpace experiments 05–07 (same `recovery-mean`, `power-law-lambda`, and `lognormal-sigma` as in `netlogo/behaviorspace_experiments.xml`). All three satisfy $\mathbb{E}[W] = 5$; the variances (legend) differ, with the heavy-tailed laws spreading mass toward longer infectious periods. The exponential curve is the continuous-time Markovian reference with rate $1/\mathbb{E}[W]$ (the implementation samples a continuous draw and rounds to ticks, as in `netlogo/virus_simulation.nlogox`).

(per-tick prevalence, seed 10001). Pass `--with-lattice-ring` to add exploratory 02–04 on Lattice4/Ring.

- Aggregation writes `output/empirical_thresholds.csv` and refreshes `report/generated_quantities.tex` unless `--no-report-macros` is set.
- Regenerate figures and `report.pdf` in one step with `python scripts/build_full_report.py` after the pipeline (requires `matplotlib`; `networkx` is optional for the network schematic).

3 Results

3.1 Spectral and mean-field references on the exported ER graph

For the primary ER instance (`expt-seed` = 10001, $N = 2000$, 5979 undirected edges on the exported list, mean degree $\langle k \rangle \approx 5.979$), Python gives $\lambda_{\max} \approx 7.181$ (see `output/lambda_max.csv`). With $\mathbb{E}[W] = 5$ ticks,

$$\beta_{\text{pred}} = \frac{1}{\lambda_{\max}\,\mathbb{E}[W]} \approx 0.0278527, \qquad \tau_{\text{pred}} = \beta_{\text{pred}}\,\mathbb{E}[W] = \frac{1}{\lambda_{\max}} \approx 0.1393.$$

We compare survival statistics to this β_{pred} on the same graph draw (`expt-seed` = 10001).

The same edge list implies a second-moment heterogeneous mean-field scale

$$\tau_{\text{het}} = \frac{\langle k \rangle}{\langle k^2 \rangle} \approx 0.24, \qquad \beta_{\text{het}} = \frac{\tau_{\text{het}}}{\mathbb{E}[W]} \approx 0.0479$$

[4]. On this instance β_{het} exceeds β_{pred} by only about 72.1%, whereas the empirical survival thresholds in Table 3 lie far above *both* references. Thus the large gap between $\hat{\beta}_{\text{surv}\,50}$ and β_{pred} is not resolved simply by switching from spectral to degree-based mean-field surrogates on the same network.

3.2 Survival curves and empirical thresholds

BehaviorSpace experiments 05–07 sweep β from 0.018 to 0.048 in steps of 0.002 with repetitions per (β, regime). Figure 3 shows the estimated fraction of runs that are not extinct at the time limit, as a function of β, for the three recovery regimes on the ER baseline. The vertical line marks β_{pred}: all three curves are already at high survival well to the right of this line, so the mean-field benchmark is conservative relative to the discrete-time ABM on this network.

Table 3 summarises $\hat{\beta}_{\text{surv}\,50}$, bootstrap uncertainty, and $\hat{\beta}/\beta_{\text{pred}}$ from `output/empirical_thresholds.csv` (primary seed 10001). Exponential and lognormal estimates in this run set sit at the largest β on the grid (0.048): the survival curve crosses 50% at or beyond the last point, so $\hat{\beta}_{\text{surv}\,50}$ is *right-censored* (ratios 1.723 and 1.723). Where the estimator sits on the boundary, the bootstrap interval often collapses to a point (Table 3, footnote). Tang power-law recovery yields a lower crossing at $\hat{\beta}_{\text{surv}\,50} = 0.044$ ($\approx 1.58\,\beta_{\text{pred}}$), the regime for which the 50% point lies strictly inside the grid here.

Figure 4 displays the same ratios relative to the spectral prediction; Figure 5 contrasts median extinction times across regimes (aggregated over β as reported in the pipeline row for each regime).

3.3 Robustness across ER draws and extinction conditioning on β

After running experiment `01` with the same seed list, each draw has its own $\lambda_{\max}$ and β_{pred} in `output/lambda_max.csv`; `scripts/aggregate_thresholds.py` merges thresholds onto BehaviorSpace rows by (`net_label`, `expt-seed`). Figure 6 compares $\hat{\beta}_{\text{surv}\,50}/\beta_{\text{pred}}$ across seeds

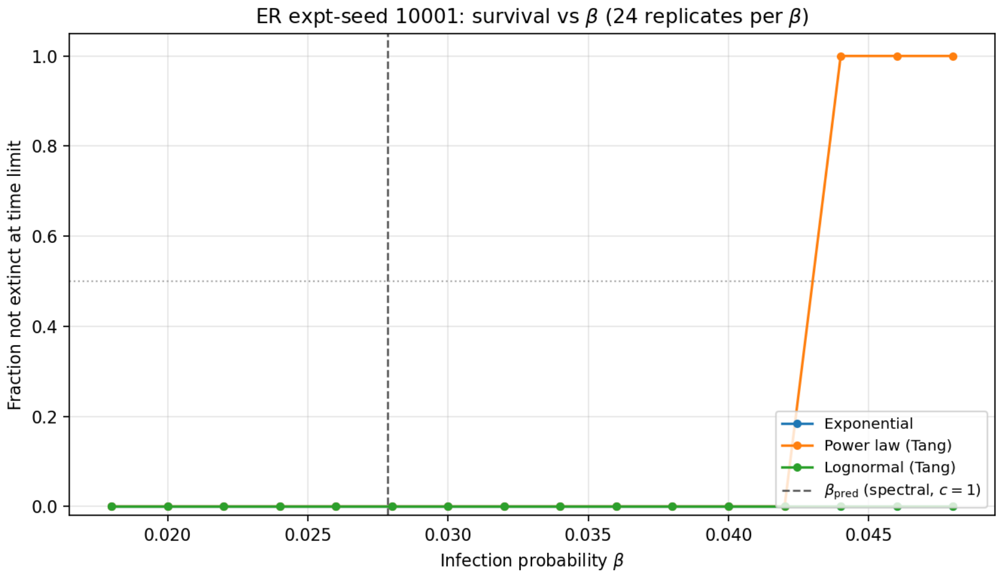

Figure 3: ER seed 10001: fraction surviving (not extinct) vs. β for the three recovery regimes. Dashed vertical line: β_{pred}. Dotted horizontal: 50% level for $\hat{\beta}_{\mathrm{surv}\,50}$. Extinction flag: `bs-out-extinct` < 0.5.

Table 3: Empirical survival threshold $\hat{\beta}_{\mathrm{surv}\,50}$ on the ER baseline (seed 10001), with $\beta_{\mathrm{pred}} = 0.0278527$. Bootstrap $n = 400$ (stratified over the β grid). Median `bs-out-final-tick` among extinct runs.

Recovery regime	$\hat{\beta}_{\mathrm{surv}\,50}$	95% CI	$\hat{\beta}/\beta_{\mathrm{pred}}$	Med. tick | extinct
Exponential	0.048	[0.048, 0.048]	1.723	8
Lognormal (Tang)	0.048	[0.048, 0.048]	1.723	9
Power law (Tang)	0.044	[0.044, 0.044]	1.58	15

Note: When $\hat{\beta}_{\mathrm{surv}\,50}$ equals the largest grid value, the threshold is censored; bootstrap realisations then often return the same boundary, yielding a degenerate interval.

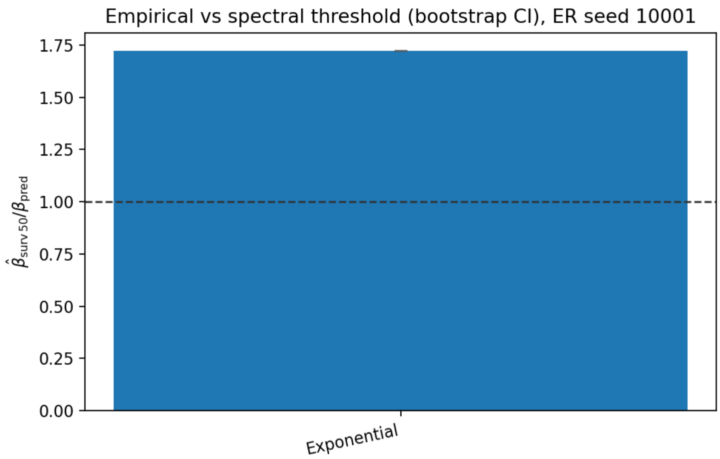

Figure 4: Ratio $\hat{\beta}_{\mathrm{surv}\,50}/\beta_{\mathrm{pred}}$ on ER with bootstrap error bars (when non-degenerate); dashed line at 1 (spectral benchmark c=1). Exponential and lognormal hit the β grid ceiling.

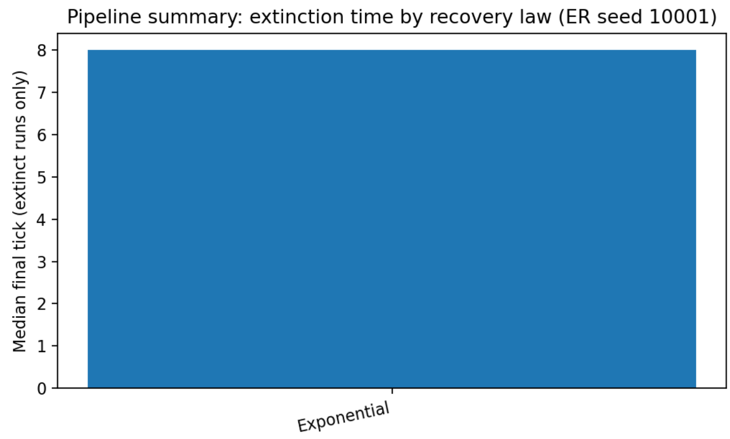

Figure 5: Median `bs-out-final-tick` among extinct runs (pipeline summary per recovery regime on ER). Power-law recovery yields substantially longer extinction times than exponential or lognormal in this summary.

(if only one seed is present in `output/empirical_thresholds.csv`, the figure script writes a placeholder reminder to re-run the multi-seed experiments). Figure 7 traces, for extinct runs only, the median final tick as a function of β with one panel per seed, separating graph noise from recovery-law effects on transients. Figure 8 shows full extinction-time distributions at four illustrative β values on the primary seed (`10001`), addressing RQ3 at finer resolution than regime-level medians.

Figure 6: Grouped comparison of $\hat{\beta}_{\mathrm{surv}\,50}/\beta_{\mathrm{pred}}$ across independent ER draws (`expt-seed` 10001–10003). After a full pipeline run with multi-seed experiments `01` and `05`–`07`, regenerate this figure with `export_report_figures.py` (project `scripts/` folder).

3.4 Microscopic vs. mean-field prevalence

Experiment `08-dynamics-trajectory-ER` records per-tick microscopic prevalence (grp-SIS) and the homogeneous mean-field SIS prevalence updated in the same model. Figure 9 shows a single replicate: the two trajectories track closely, while stochastic extinction can end the

Figure 7: Median `bs-out-final-tick` among runs that hit absorption before the horizon, vs. β, faceted by `expt-seed`.

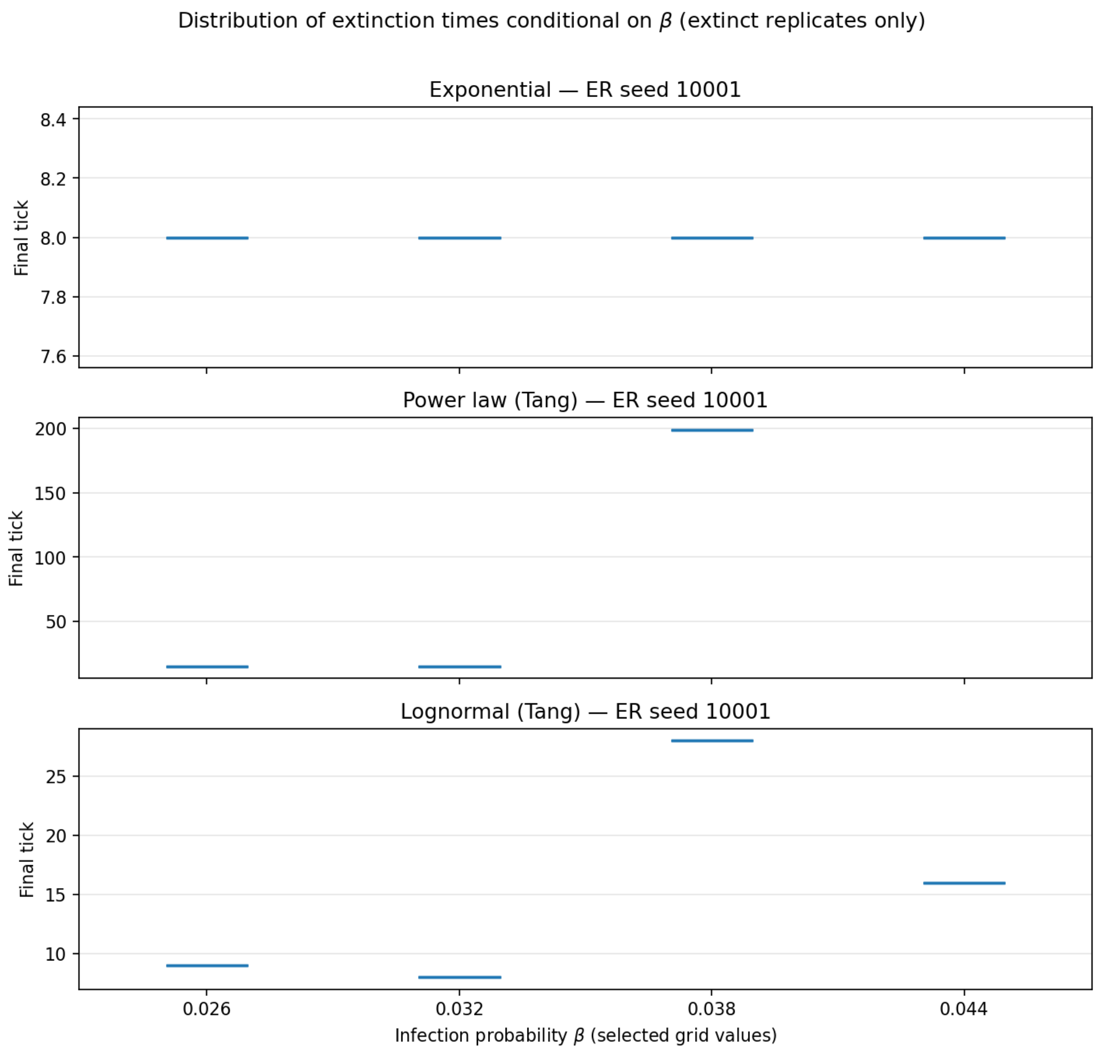

Figure 8: Violin plots of extinction times (extinct runs only) at $\beta \in \{0.026, 0.032, 0.038, 0.044\}$, ER seed 10001.

microscopic run early (exit when no infected remain).

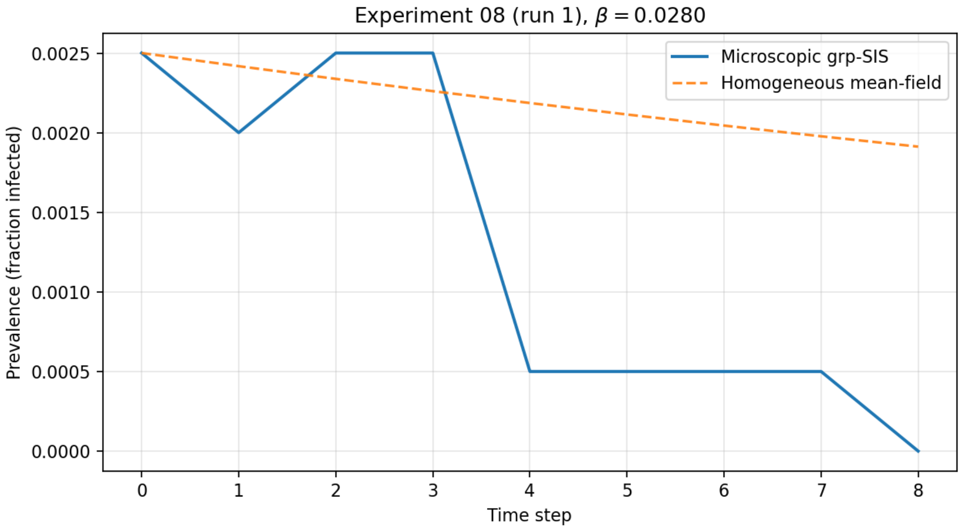

Figure 9: One BehaviorSpace run from `08-dynamics-trajectory-ER`: microscopic prevalence vs. NetLogo mean-field SIS curve at the same parameters.

4 Discussion

Spectral benchmark vs. ABM survival (RQ1). The empirical $\hat{\beta}_{\mathrm{surv}\,50}$ lies clearly above β_{pred}: in τ-space, $\hat{\tau}_{50} = \hat{\beta}_{\mathrm{surv}\,50}\mathbb{E}[W]$ corresponds to an effective multiplier $c \approx \hat{\tau}_{50}\,\lambda_{\max}$ of about 1.58–1.723 on our grid (seed 10001), not c=1. So the mean-field spectral line is best read as a *scale* or optimistic bound, not a sharp critical value, for this discrete-time, finite-N, agent-based grp-SIS implementation. Finite-size effects, discretisation, and spatial correlations beyond mean field are the usual explanations; the trajectory check (Figure 9) shows that the *homogeneous* mean-field equation itself matches the simulation when both are defined on the same state variables, so the gap in Table 3 is about *where survival becomes likely* in the stochastic model, not about a coding inconsistency between ODE and micro-dynamics.

Recovery shape (RQ2) and dynamics (RQ3). The original hypothesis emphasised easier persistence under heavy tails at fixed $\mathbb{E}[W]$. On this ER instance and β grid, we do not see a uniform lowering of $\hat{\beta}_{\mathrm{surv}\,50}$ for *both* heavy-tailed regimes: lognormal matches exponential at the grid cap, while power-law recovery crosses 50% at a slightly lower β. Thus RQ2 receives a nuanced answer—possible ordering effects between specific heavy-tailed laws rather than a single "heavy tail" shift. For RQ3, the median extinction-time summary (Figure 5) supports the view that recovery shape reshapes *transient* and extinction behaviour more visibly than a shared spectral β_{pred} might suggest: Tang power-law recovery is associated with longer extinction times in the pipeline summary. A fuller answer would examine extinction-time *distributions* by β (not only the aggregated median) and persistence of late-window prevalence.

Risks to validity and mitigation. Several design choices could mislead interpretation if left implicit; the mitigations below address some but not all limitations. *(i) Finite horizon.* Survival is defined at `max-ticks` = 10,000, so a run classified "surviving" may still die out later; longer horizons or absorption-time analysis would reduce this ambiguity, at higher cost. *(ii) Grid and censoring.* $\hat{\beta}_{\mathrm{surv}\,50}$ lives on a coarse $\Delta\beta = 0.002$ grid and hits the upper cap for two regimes here; extending the sweep, adaptive refinement, or a logistic fit to survival vs. β would narrow discretisation and boundary bias. *(iii) Initial seed.* Fixing I_0=5 targets a small outbreak; larger I_0 can shift finite-N extinction probabilities and thus survival rates even when the infinite-population threshold is unchanged—we hold I_0 constant across laws but do not claim invariance to I_0. *(iv) Replication.* Twenty-four replicates per (β, regime) stabilise the survival fraction estimator; bootstrap intervals quantify Monte Carlo noise conditional on the graph draw, not graph ensemble uncertainty (addressed partly by multiple `expt-seed` values). *(v) Discrete time vs. continuous theory.* The spectral line is a continuous-time mean-field benchmark; Section 2 already notes the discrete-time mapping—treating disagreement as diagnostic of that gap is deliberate, not an implementation error.

Limitations. Estimators that hit the largest β are censored: extending the β grid or fitting a dose–response to survival vs. β would sharpen $\hat{\beta}_{\text{surv}\,50}$ for exponential and lognormal curves. The design now includes three ER draws (`expt-seed` 10001–10003); Figures 6–7 summarise cross-graph variation once those runs are available (Table 3 remains focused on seed 10001 for continuity with the main text). Figure 8 illustrates extinction-time *distributions* at selected β but is not exhaustive across the full grid. The default pipeline excludes lattice/ring experiments `02–04`; the narrative focuses on the ER baseline aligned with `output/edges/edges_ER_10001.csv` for the primary spectral reference. The study concerns synthetic disease transmission only; no human or animal data are involved.

5 Conclusion

For grp-SIS on a moderately dense ER graph with fixed mean recovery time, the spectral benchmark $\beta_{\text{pred}} = 1/(\lambda_{\max}\mathbb{E}[W])$ remains a useful scale but systematically underestimates the β at which survival at the fixed horizon becomes common in this ABM: in τ-space, $\hat{\tau}_{50}/\tau_{\text{pred}}$ is about 1.58–1.723 on our grid for seed 10001, not c=1. A standard heterogeneous mean-field β from degree moments moves only slightly relative to β_{pred} and does not absorb that gap. Differences between recovery *laws* show up in threshold ordering at finite resolution (power law vs. exponential/lognormal on this sweep) and more clearly in extinction-time summaries. Overall, heavy-tailed recovery does not uniformly lower the empirical survival threshold relative to Markovian recovery here; it interacts with the chosen law and with censoring. Extending β sweeps, adding graph ensembles, and analysing extinction-time distributions conditional on β would further connect Tang-style non-Markovian recovery to spectral threshold practice.

Data and code availability

BehaviorSpace table exports, edge lists, and aggregated threshold CSVs are under `output/` in the project repository. The NetLogo model and experiment definitions are in `netlogo/virus_simulation.nlogox` and `netlogo/behaviorspace_experiments.xml`. Figures in this PDF were generated with `scripts/export_report_figures.py`.

References

[1] P. Erdős and A. Rényi. On random graphs. *Publ. Math. Debrecen* **6**, 290–297 (1959).

[2] E. Cator, R. van de Bovenkamp, and P. Van Mieghem. Susceptible–infected–susceptible epidemics on networks with general infection and cure times. *Phys. Rev. E* **87**, 062816 (2013).

[3] D. Chakrabarti, Y. Wang, C. Wang, J. Leskovec, and C. Faloutsos. Epidemic thresholds in real networks. *ACM Trans. Inf. Syst. Secur.* **10**, 13 (2008).

[4] R. Pastor-Satorras, C. Castellano, P. Van Mieghem, and A. Vespignani. Epidemic processes in complex networks. *Rev. Mod. Phys.* **87**, 925–979 (2015).

[5] J. Tang, Y. Yao, M. Xie, and M. Feng. SIS epidemic modelling on homogeneous networked system: General recovering process and mean-field perspective. *Appl. Math. Modelling* (2025); preprint arXiv:2505.12290v2.

[6] P. Van Mieghem. The N-intertwined SIS epidemic network model. *Computing* **93**, 147–169 (2011).

A BehaviorSpace experiment configurations (NetLogo)

This appendix documents the NetLogo BehaviorSpace experiments as defined in `netlogo/behaviorspace_experiments.xml` (and mirrored into `netlogo/virus_simulation.nlogox` for headless execution). The **default** headless pipeline (`scripts/run_pipeline.py--all`) runs the Erdős–Rényi block `01`, `05`–`08` only; experiments `02`–`04` are retained for optional lattice/ring exploration (`--with-lattice-ring`).

A.1 Common elements

- **Model**: `netlogo/virus_simulation.nlogox` (NetLogo 7.x).
- **Exit condition (threshold experiments)**: runs stop at the time horizon or absorption,

$$(\texttt{ticks} \geq \texttt{sim-tick-limit}) \quad \text{or} \quad (\text{no infected agents}).$$

- **Key controlled settings (reported below)**: network specification (`interconnection-structure`, `network-type`, `expt-seed`, `num-nodes`, `avg-degree`), epidemic parameters (`infection-prob`, `recovery-mean`, recovery distribution chooser and its parameters), initial condition (`initial-infected`), and horizon (`max-ticks`).

A.2 Experiment 01: network export

`01-export-networks`

- **Purpose**: export edge lists for subsequent computation of $\lambda_{\max}$ and mean-field reference quantities.
- **Repetitions**: 1.
- **Setup (summary)**: disables collection of epidemic metrics; calls `export-network-edges` to write `output/edges/edges_\{net\}\_\{seed\}.csv`.
- **Exit condition**: `true` (immediate after export).
- **Metrics**: `safe-net-label`, `count links`, `num-nodes`, `bs-out-mean-degree`.
- **Constants and sweeps**:
  - `interconnection-structure` = `"other"` (Erdős–Rényi via `network-type`).
  - `network-type` = `"Erdos-Renyi (random)"`.
  - `expt-seed` $\in \{10001, 10002, 10003\}$.
  - `num-nodes` = 2000.
  - `avg-degree` = 6; `rewiring-prob` = 0.07; `initial-infected` = 5; `max-ticks` = 3000.

A.3 Experiments 02–04: optional coarse sweeps on regular structures

Not run in the default ER-focused pipeline; include with `--with-lattice-ring`.

`02-threshold-exponential`

- **Repetitions**: 8.
- **Recovery law**: `"exponential"`.
- **Metrics**: `bs-out-extinct`, `bs-out-final-tick`, `bs-out-late-mean-prevalence`, `bs-out-tau-sim`, `bs-out-mean-degree`.
- **Constants and sweeps**: `interconnection-structure` $\in$ {`"Lattice4"`, `"Ring"`}; `expt-seed`=1001; `num-nodes`=200; `avg-degree`=6; `initial-infected`=5; `recovery-mean`=5; `max-ticks`=3000; stepped `infection-prob` from 0.02 to 0.56 in steps of 0.03.

`03-threshold-power-law-Tang`

- **Repetitions**: 8.
- **Recovery law**: `"power law (Tang)"` with `power-law-lambda`=4.24.
- **Metrics**: same as experiment 02.
- **Constants and sweeps**: same as experiment 02, with the additional constant `power-law-lambda`=4.24.

`04-threshold-lognormal-Tang`

- **Repetitions**: 8.
- **Recovery law**: `"lognormal (Tang)"` with `lognormal-sigma`=1.
- **Metrics**: same as experiment 02.
- **Constants and sweeps**: same as experiment 02, with the additional constant `lognormal-sigma`=1.

A.4 Experiments 05–07: ER baseline experiments (main results)

`05-baseline-empirical-threshold-ER`

- **Repetitions**: 24.
- **Network**: `interconnection-structure="other"`, `network-type="Erdos-Renyi (random)"`; `expt-seed` $\in \{10001, 10002, 10003\}$; `num-nodes`=2000; `avg-degree`=6.
- **Recovery law**: `"exponential"` with `recovery-mean`=5.
- **Horizon**: `max-ticks`=10000.
- **Metrics**: `bs-out-extinct`, `bs-out-final-tick`, `bs-out-late-mean-prevalence`, `bs-out-tau-sim`, `bs-out-mean-degree`.
- **Sweep**: stepped `infection-prob` from 0.018 to 0.048 in steps of 0.002.

`06-baseline-ER-power-law-Tang`

- **Repetitions**: 24 (same `expt-seed` list as experiment 05; one ER draw per seed).
- **Recovery law**: `"power law (Tang)"` with `recovery-mean`=5 and `power-law-lambda`=4.24.
- **Horizon, metrics, sweep**: identical to experiment 05.

`07-baseline-ER-lognormal-Tang`

- **Repetitions**: 24 (same `expt-seed` list as experiment 05; one ER draw per seed).
- **Recovery law**: `"lognormal (Tang)"` with `recovery-mean`=5 and `lognormal-sigma`=1.
- **Horizon, metrics, sweep**: identical to experiment 05.

A.5 Experiment 08: prevalence trajectories (grp-SIS vs mean-field)

`08-dynamics-trajectory-ER`

- **Purpose**: export per-tick prevalence for the microscopic grp-SIS simulation and the in-model homogeneous mean-field SIS curve for the same parameters (part of the default ER pipeline together with `01` and `05`–`07`).
- **Repetitions**: 5.
- **RunMetricsEveryStep**: `true` (per-tick table export).
- **Network and initial condition**: ER baseline with `expt-seed`=10001; `num-nodes`=2000; `avg-degree`=6; `initial-infected`=5.
- **Recovery law**: `"exponential"` with `recovery-mean`=5.
- **Horizon**: `max-ticks`=2000.
- **Infection strength**: `infection-prob`=0.028 (single value).
- **Metrics**: `bs-out-prev-grp`, `bs-out-prev-mf`, `bs-out-count-S`, `bs-out-count-I`, `bs-out-mean-degree`.